

CACE

Circuit Automatic Characterization Engine

Leo Moser & Tim Edwards

Efabless

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You found a nice open-source analog IP block:

- What is the structure of the repository?
- Is the given information accurate/up-to-date?
- How to run simulations and verify certain parameters?
- Is the repository even in a complete state?

We would like to have:

- A structure around analog design to ensure reusability.
- Projects that use an open PDK and are publicly hosted should be fully verifiable.
- Preferably with the least effort...

CACE is a tool for allowing analog and mixed-signal circuits to be fully characterized across corners and with Monte Carlo and mismatch analysis using open source tools, in a way that is formalized enough to allow any circuit pulled from a public repository to be both verified and reused.

<https://github.com/efabless/cace>

The screenshot displays the GitHub repository interface for `efabless/cace`. At the top, the URL `https://github.com/efabless/cace` is shown. Below the repository name, there's a navigation bar with tabs for 'main', '1 branch', and '2 tags'. A search bar and 'Add file' button are also present. The main content area lists files and folders with their last commit dates. A merge pull request is visible at the top. On the right, there's an 'About' section with a description of the project and a 'Releases' section. The bottom part of the image shows the README content, which includes the project title 'CACE Circuit Automatic Characterization Engine', a list of supported tools (Apple FTL, JTAG, JTAG, JTAG, JTAG, JTAG), and a brief description of the project's purpose.

File/Folder	Last Commit
<code>.github</code>	Add .nix configs
<code>cace</code>	Abort early if netlist generation fails
<code>docs</code>	Docs fixup
<code>nix</code>	Add .nix configs
<code>tests</code>	Apply blue
<code>.gitignore</code>	Add .gitignore
<code>.readthedocs.yaml</code>	Update path to config
<code>ChangeLog.md</code>	Abort early if netlist generation fails
<code>LICENS</code>	Seeded repository with source files
<code>Makefile</code>	Improve and clarify the docs
<code>README.md</code>	Multiprocessing rewrite
<code>default.nix</code>	Add progress bar and render Markdown
<code>flake.lock</code>	Various files
<code>flake.nix</code>	Flake
<code>pyproject.toml</code>	Lower setup tools, some requirements to >=7
<code>requirements.txt</code>	Add progress bar and render Markdown
<code>requirements_dev.txt</code>	Integrate blue
<code>requirements_docs.txt</code>	Improve and clarify the docs
<code>shell.nix</code>	Add .nix configs

README Apache-2.0 license

CACE

Circuit Automatic Characterization Engine

License: Apple FTL, JTAG, JTAG, JTAG, JTAG, JTAG

CACE is a set of python scripts that take an input file in the CACE 4.0 format and uses the information found there in combination with CACE-compatible testbenches and analysis scripts to characterize a circuit and to produce a datasheet showing the circuit performance.

- 1. BYOT - Bring Your Own Tools
 - Install CACE via pip: `pip3 install --upgrade cace`
 - Make sure all other tools are installed
 - Works for Docker containers
- 2. Nix
 - CACE provides a Nix configuration
 - Setup Nix & binary cache
 - Create a shell: `nix-shell`

<https://cace.readthedocs.io/>

- Installation Procedure
- Datasheet Format
- Template Format
- Tutorial coming soonTM

CACE Documentation

🔍 Search

Getting Started

Usage

Reference Manual

Tutorial

Examples

Development

The Documentation for CACE



ON THIS PAGE

Circuit Automatic Characterization
Engine

Circuit Automatic Characterization Engine

CACE is a framework for analog and mixed-signal circuits that enables automatic characterization under various conditions and with Monte Carlo and mismatch analysis. After all parameters have been run under the given conditions, CACE produces a summary showing the circuit performance. CACE requires the designer to create a specification, called the *datasheet* in CACE jargon, which contains the parameters, their conditions and limits. In addition, the designer must set up the directory structure of the design in a regular way and create template testbenches for the parameters.

Setting up CACE requires some additional effort, but the benefits of using CACE are clear.

- **Specification and project structure** Each circuit design requires a datasheet that serves both as documentation for the specification and also as input for CACE. Designs must adhere to a regular directory structure, thus other designs using CACE feel familiar.
- **Reproducibility and re-use** Designs that have the CACE system set up can be fully and automatically characterized. This makes it easy to verify the correctness and completeness of analog circuits, facilitating the reuse of designs.
- **Good design practices** Finally, CACE encourages good analog design practices, fostering trust in open source analog design.

Follow the navigation element below (or check the sidebar on the left) to get started.

- Efabless “Open Galaxy” platform
- First design challenge: bandgap in X-Fab XH035
- A lot has happened in between...
- Now CACE is modernized and freely available!

CACE Study: 5-Transistor OTA

- Simple example
- How to setup CACE
- Continuous integration

<https://github.com/mole99/ota-5t>

ota-5t Public

Pin Unwatch 1 Fork 0 Star 0

main 1 Branch 0 Tags

Go to file Add file Code

Commit History

mole99	Enable debug output	e738e78 · 4 hours ago	31 Commits
.github/workflows	Enable debug output	4 hours ago	
cace	Update README	6 hours ago	
gds	Add layout with static pcells	yesterday	
xschem	Update gitignore	8 hours ago	
.gitignore	Update gitignore	8 hours ago	
LICENSE	Initial commit	2 weeks ago	
README.md	Update README	6 hours ago	

README Apache-2.0 license

A simple 5-transistor OTA

This example showcases how to setup a project for [CACE](#).

About

A simple 5-transistor OTA.

- Readme
- Apache-2.0 license
- Activity
- 0 stars
- 1 watching
- 0 forks

Releases

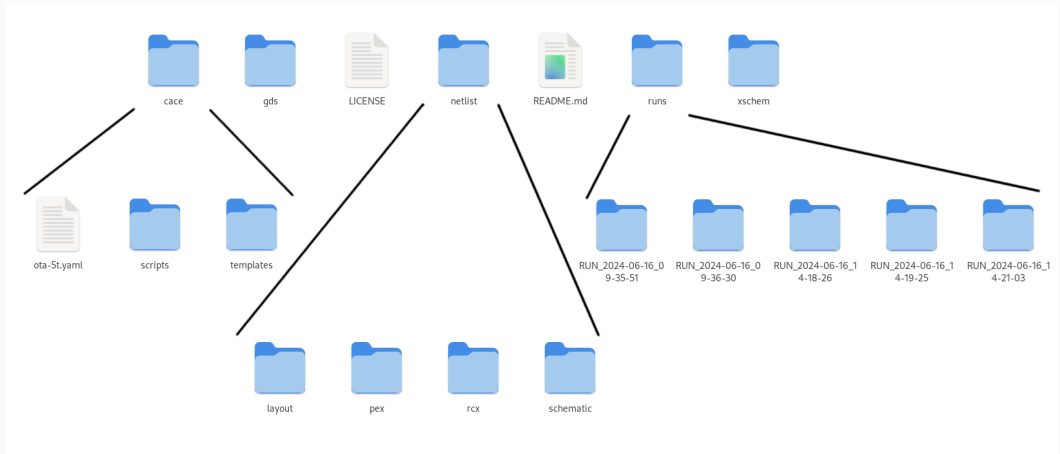
No releases published
[Create a new release](#)

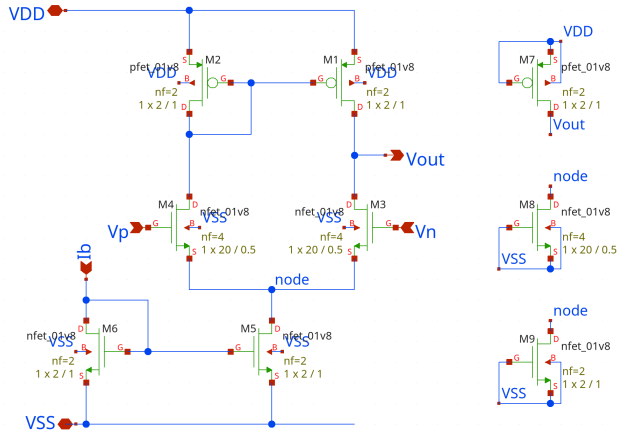
Packages

No packages published
[Publish your first package](#)

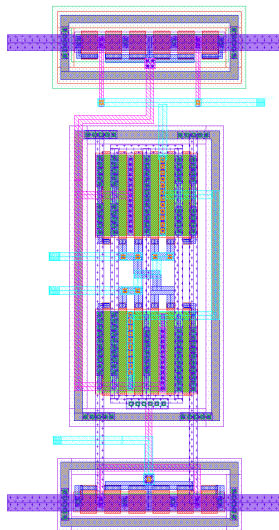
Languages

Tcl 59.9% Python 40.1%





- KLayout PCells
- Matching techniques:
 - Interdigitized layout
 - Common centroid layout
 - Dummy transistors



- Information about:
 - Design
 - Author
 - License
 - Paths
 - Dependencies...

```
1 #-----
2 # CACE circuit characterization file
3 #-----
4
5 name:          ota-5t
6 description:   A simple 5-transistor OTA
7 commit:       n/a
8 PDK:          sky130A
9
10 cace_format:   5.0
11
12 authorship:
13   designer:    Leo Moser
14   company:     Efabless
15   creation_date: May 27, 2024
16   license:     Apache 2.0
17
18 paths:
19   root:        ..
20   schematic:   xschem
21   layout:      gds
22   netlist:     netlist
23   documentation: doc
24
```

- Pin information
- Checked against schematic
- Optional

```
25 pins:
26   VDD:
27     description: Positive analog power supply
28     type: power
29     direction: inout
30     Vmin: 1.7
31     Vmax: 1.9
32   VSS:
33     description: Analog ground
34     type: ground
35     direction: inout
36   Ib:
37     description: Bias current input
38     type: signal
39     direction: input
40   Vp:
41     description: Voltage positive input
42     type: signal
43     direction: input
44   Vn:
45     description: Voltage negative input
46     type: signal
47     direction: input
48   Vout:
49     description: Voltage output
50     type: signal
51     direction: output
52
```

- Description, values and units of the default conditions

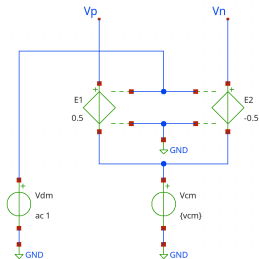
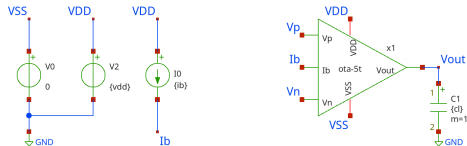
```
53 default_conditions:
54   vdd:
55     description: Analog power supply voltage
56     display: Vdd
57     unit: V
58     typical: 1.8
59   vcm:
60     description: Input common mode voltage
61     display: Vcm
62     unit: V
63     typical: 0.9
64   ib:
65     description: Bias current
66     display: Ib
67     unit: uA
68     typical: 10
69   cl:
70     description: Output load capacitance
71     display: CLoad
72     unit: pF
73     maximum: 1
74   corner:
75     description: Process corner
76     display: Corner
77     typical: tt
78   temperature:
79     description: Ambient temperature
80     display: Temp
81     unit: °C
82     typical: 27
83
```

- Configuration for simulations
- Specify:
 - Limits
 - Settings
 - Conditions
 - Variables
 - Measure
 - Plots...

```
84 electrical_parameters:
85   A0:
86     description: DC gain
87     display: A0
88     unit: V/V
89     spec:
90       minimum:
91         value: 50
92         fail: true
93       typical:
94         value: any
95       maximum:
96         value: any
97     simulate:
98       ngspice:
99         template: ota-5t_tb.spice
100        format: ascii
101        suffix: .data
102        variables: [result, 'null', 'null']
103      conditions:
104        vdd:
105          minimum: 1.7
106          typical: 1.8
107          maximum: 1.9
108        corner:
109          enumerate: [tt, ff, ss] # fs, sf
110      temperature:
111        minimum: -40
112        typical: 27
113        maximum: 130
114
```

- Properties of the layout
- Area, DRC, LVS...
- Often maximum: 0

```
175 physical_parameters:
176   area:
177     description: Total circuit layout area
178     display: Area
179     unit:  $\mu\text{m}^2$ 
180     spec:
181       maximum:
182         value: 600
183         fail: true
184     evaluate: cace_area
185
186   LVS_errors:
187     description: LVS errors
188     display: LVS errors
189     spec:
190       maximum:
191         value: 0
192         fail: true
193     evaluate:
194       cace_lvs:
195         script: run_project_lvs.tcl
196
197   DRC_errors:
198     description: DRC errors
199     display: DRC errors
200     spec:
201       maximum:
202         value: 0
203         fail: true
204     evaluate: cace_drc
```



SETUP

```
.lib {PDK_ROOT}/{PDK}/libs.tech/combined/sky130.lib.spice {corner}
.include {DUT_path}
.temp {temperature}
```

CONTROL

```
.control
* run ac simulation
ac dec 20 1k 100e6

* measure parameters
let vout_mag = abs(v(Vout))
let vout_phase_margin = phase(v(Vout)) * 180/pi + 180
meas ac A0 find vout_mag at=1k
meas ac UGF when vout_mag=1 fall=1
meas ac PM find vout_phase_margin when vout_mag=1

echo $A0 $UGF $PM > {simpath}/{filename}_{N}.data
quit
.endc
```

Just run CACE:

```
$ cace
```

Choose the netlist source:

```
$ cace --source rcx --parameter A0 PM
```

Debugging:

```
$ cace --source rcx --log-level DEBUG \  
  --parallel_parameters 1 --sequential
```

GUI:

```
$ cace-gui
```

```

leo@debian-pc: ~/Projects/cace_upstream
[16:21:02] INFO Parameter LVS_errors: Completed
[16:21:02] INFO Parameter DRC_errors: Started
[16:21:02] INFO Parameter DRC_errors: Evaluating physical parameter
[16:21:02] INFO Checking physical parameter DRC_errors
[16:21:02] INFO Running magic to get layout DRC report.
[16:21:03] INFO Parameter DRC_errors: Completed
[16:21:05] INFO Parameter UGF: Starting task with id 0
[16:21:05] INFO Parameter UGF: Starting task with id 1
[16:21:05] INFO Parameter UGF: Starting task with id 2
[16:21:05] INFO Parameter UGF: Starting task with id 3
[16:21:05] INFO Parameter UGF: Starting task with id 4
[16:21:05] INFO Parameter UGF: Logging to 'runs/RUN_2024-06-16_16-21-01/parameters/UGF/UGF_1.log'...
[16:21:05] INFO Parameter UGF: Logging to 'runs/RUN_2024-06-16_16-21-01/parameters/UGF/UGF_2.log'...
[16:21:05] INFO Parameter A0: Starting task with id 0
[16:21:05] INFO Parameter UGF: Logging to 'runs/RUN_2024-06-16_16-21-01/parameters/UGF/UGF_3.log'...
[16:21:05] INFO Parameter UGF: Starting task with id 5
[16:21:05] INFO Parameter PM: Starting task with id 0
[16:21:05] INFO Parameter UGF: Logging to 'runs/RUN_2024-06-16_16-21-01/parameters/UGF/UGF_4.log'...
[16:21:05] INFO Parameter A0: Starting task with id 1
[16:21:05] INFO Parameter UGF: Logging to 'runs/RUN_2024-06-16_16-21-01/parameters/UGF/UGF_5.log'...
[16:21:05] INFO Parameter UGF: Starting task with id 6
[16:21:05] INFO Parameter PM: Starting task with id 1
[16:21:05] INFO Parameter A0: Logging to 'runs/RUN_2024-06-16_16-21-01/parameters/A0/A0_1.log'...
[16:21:05] INFO Parameter PM: Logging to 'runs/RUN_2024-06-16_16-21-01/parameters/PM/PM_1.log'...
[16:21:05] INFO Parameter A0: Starting task with id 2
[16:21:05] INFO Parameter UGF: Logging to 'runs/RUN_2024-06-16_16-21-01/parameters/UGF/UGF_6.log'...
[16:21:05] INFO Parameter A0: Logging to 'runs/RUN_2024-06-16_16-21-01/parameters/A0/A0_2.log'...
[16:21:05] INFO Parameter UGF: Starting task with id 7
[16:21:05] INFO Parameter PM: Starting task with id 2
[16:21:05] INFO Parameter A0: Starting task with id 3
[16:21:05] INFO Parameter UGF: Logging to 'runs/RUN_2024-06-16_16-21-01/parameters/UGF/UGF_7.log'...
[16:21:05] INFO Parameter PM: Logging to 'runs/RUN_2024-06-16_16-21-01/parameters/PM/PM_2.log'...
[16:21:05] INFO Parameter A0: Logging to 'runs/RUN_2024-06-16_16-21-01/parameters/A0/A0_3.log'...
[16:21:05] INFO Parameter UGF: Starting task with id 8
[16:21:05] INFO Parameter PM: Starting task with id 3
[16:21:05] INFO Parameter A0: Starting task with id 4
[16:21:05] INFO Parameter A0: Logging to 'runs/RUN_2024-06-16_16-21-01/parameters/A0/A0_4.log'...
[16:21:05] INFO Parameter PM: Logging to 'runs/RUN_2024-06-16_16-21-01/parameters/PM/PM_3.log'...
[16:21:05] INFO Parameter UGF: Logging to 'runs/RUN_2024-06-16_16-21-01/parameters/UGF/UGF_8.log'...
[16:21:05] INFO Parameter UGF: Starting task with id 9
[16:21:05] INFO Parameter PM: Starting task with id 4

Running Parameters 3/6 0:00:04
UGF 0/27 0:00:00
A0 0/27 0:00:00
PM 0/27 0:00:00

```

19 / 34

```

leo@debian-pc: ~/Projects/cace_upstream
[16:21:54] INFO Done with CACE simulations and evaluations. cace_cli.py:303

CACE Summary

general
* name: ota-5t
* description: A simple 5-transistor OTA
* commit: n/a
* PDK: sky130A
* cace_format: 5.0

authorship
* designer: Leo Moser
* company: Efabless
* creation_date: May 27, 2024
* license: Apache 2.0

netlist source: best

Electrical Parameters



| Parameter | Testbench       | Min Limit | Min Value    | Typ Target | Typ Value    | Max Limit | Max Value    | Status |
|-----------|-----------------|-----------|--------------|------------|--------------|-----------|--------------|--------|
| A0        | ota-5t_tb.spice | 50 V/V    | 58.55 V/V    | any        | 81.73 V/V    | any       | 99.36 V/V    | Pass ✓ |
| UGF       | ota-5t_tb.spice | 1e6 Hz    | 1.249e+07 Hz | any        | 1.565e+07 Hz | any       | 1.887e+07 Hz | Pass ✓ |
| PM        | ota-5t_tb.spice | 60 °      | 85.98 °      | any        | 86.31 °      | any       | 86.94 °      | Pass ✓ |



Physical Parameters



| Parameter  | Tool      | Min Limit | Min Value | Typ Target | Typ Value | Max Limit | Max Value | Status |
|------------|-----------|-----------|-----------|------------|-----------|-----------|-----------|--------|
| Area       | cace_area |           |           |            |           | 600 μm²   | 547 μm²   | Pass ✓ |
| LVS errors | cace_lvs  |           |           |            |           |           | 0         | Pass ✓ |
| DRC errors | cace_drc  |           |           |            |           |           | 0         | Pass ✓ |



[nix-shell:~/Nextcloud/Efabless/FSiC/ota-5t]$

```

CACE

CACE Datasheet: ota-5t.yaml cace/ota-5t.yaml Netlist from: Schematic Capture

Project IP name: ota-5t sky130A A simple 5-transistor OTA

Parameter	Testbench	Min	Typ	Max	Status	Simulate All
A0	ota-5t_tb.spice	50 V/V	58.84 V/V	(no target) 81.98 V/V	(no limit) 99.59 V/V	pass Simulate
UGF	ota-5t_tb.spice	1e6 Hz		(no target)	(no limit)	(not checked) Simulate
PM	ota-5t_tb.spice	60 °	87.01 °	(no target) 87.28 °	(no limit) 87.77 °	pass Simulate
Area estimate	cace_area	(no limit)		(no target)	600 μm²	184 μm² pass Check
Invalid device check	cace_lvs	(no limit)		(no target)	0	(not checked) Check
DRC errors	cace_drc	(no limit)		(no target)	0	(N/A) Check

```
[16:23:29] INFO Parameter PM: Logging to 'runs/RUN_2024-06-16_16-23-03/parameters/PM/PM_21.log'-
simulation_job.py:343
[16:23:30] INFO Parameter PM: Logging to 'runs/RUN_2024-06-16_16-23-03/parameters/PM/PM_22.log'-
simulation_job.py:343
[16:23:30] INFO Parameter PM: Logging to 'runs/RUN_2024-06-16_16-23-03/parameters/PM/PM_23.log'-
simulation_job.py:343
[16:23:30] INFO Parameter PM: Logging to 'runs/RUN_2024-06-16_16-23-03/parameters/PM/PM_24.log'-
simulation_job.py:343
[16:23:30] INFO Parameter PM: Logging to 'runs/RUN_2024-06-16_16-23-03/parameters/PM/PM_25.log'-
simulation_job.py:343
[16:23:30] INFO Parameter PM: Logging to 'runs/RUN_2024-06-16_16-23-03/parameters/PM/PM_26.log'-
simulation_job.py:343
[16:23:30] INFO Parameter PM: Logging to 'runs/RUN_2024-06-16_16-23-03/parameters/PM/PM_27.log'-
simulation_job.py:343
[16:23:31] INFO Simulation of A0 has completed.
cace_gui.py:430
[16:23:31] INFO Parameter A0: Completed
parameter.py:135
[16:23:35] INFO Simulation of PM has completed.
cace_gui.py:430
[16:23:35] INFO Parameter PM: Completed
parameter.py:135
```

Quit Save Save As Load HTML Help Settings

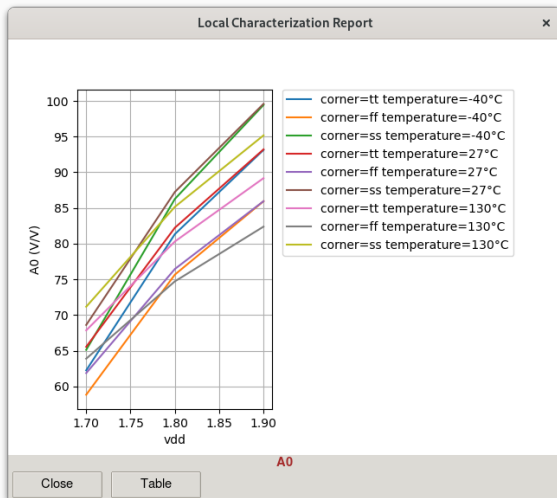
Local Characterization Report

Electrical Parameter: A0 Testbench: ota-5t_ib.spice Min Limit: 50 f/V V/V Max Limit: any V/V

Constant Conditions: ib = 10.0 uA cl = 1.0 pF vcm = 0.9 V

result (f)	vdd	corner	temperature	testbench
V/V	V		°C	
(1.7 1.9)	(tt ff ss)	(-40.0 130.0)	(1 27)	
58.8366	1.7	ff	-40	A0_4.spice
61.8698	1.7	ff	27	A0_13.spice
62.2504	1.7	tt	-40	A0_1.spice
63.8884	1.7	ff	130	A0_22.spice
65.1241	1.7	ss	-40	A0_7.spice
65.5546	1.7	tt	27	A0_10.spice
67.8556	1.7	tt	130	A0_19.spice
68.6125	1.7	ss	27	A0_16.spice
71.1685	1.7	ss	130	A0_25.spice
74.7233	1.8	ff	130	A0_23.spice
75.6682	1.8	ff	-40	A0_5.spice
76.4545	1.8	ff	27	A0_14.spice
80.3086	1.8	tt	130	A0_20.spice
81.3413	1.8	tt	-40	A0_2.spice
82.2354	1.8	tt	27	A0_11.spice
82.3911	1.9	ff	130	A0_24.spice
85.1625	1.8	ss	130	A0_26.spice
85.9473	1.9	ff	-40	A0_6.spice
85.959	1.9	ff	27	A0_15.spice
86.2689	1.8	ss	-40	A0_8.spice
87.2486	1.8	ss	27	A0_17.spice
89.1815	1.9	tt	130	A0_21.spice
93.1377	1.9	tt	-40	A0_3.spice
93.2129	1.9	tt	27	A0_12.spice
95.1768	1.9	ss	130	A0_27.spice
99.4403	1.9	ss	-40	A0_9.spice
99.5943	1.9	ss	27	A0_18.spice

Close



Continuous Integration

How do you ensure that your design is **always reproducible** and in a **working state**?

- Perform the entire characterization with every change!
- We provide a GitHub Action for CACE
- Uses the same Nix environment as locally

<https://github.com/efabless/cace-action>

cace-action Public

1 main 1 Branch 0 Tags

Go to file t Add file + Code +

Commit	Message	Time	Comments
male99	Update CACE	6523401 · 1 hour ago	5 Comments
	LICENSE	Initial commit	2 weeks ago
	README.md	Update README.md	4 hours ago
	action.yml	Update CACE	1 hour ago

README Apache-2.0 license

cace-action

GitHub composite action for CACE.

Insert the composite action into a workflow to set up the environment (PDK and tools) for CACE and execute it inside your repository:

```
...
- name: Setup Environment and Run CACE
  uses: efabless/cace-action@main
  with:
    token: ${{ secrets.GITHUB_TOKEN }}
...
```

The following inputs are provided to customize the action, for example:

```
...
- name: Setup Environment and Run CACE
  uses: efabless/cace-action@main
  with:
    pdk_family: 'sky130'
    open_pdk_rev: '495af10b7ee4d0b799566a7903b622aee690ac9'
    cace_rev: '883e0d0e7ab0429421d088e6f0a0e6e608acdb'
    cace_root: '.'
    cace_dataSheet: 'cace/ota-st_yml'
    cace_source: 'schematic'
    cace_args: '--log-level 00800'
    token: ${{ secrets.GITHUB_TOKEN }}
...
```

See `action.yml` for default values.

About

GitHub composite action for CACE

- Readme
- Apache-2.0 license
- Activity
- Custom properties
- Stars
- Watching
- Forks
- Report repository

Releases

No releases published
[Create a new release](#)

Packages

No packages published
[Publish your first package](#)

- Run action on events
- Set up the environment
- Run CACE
- Optionally:
 - Upload artifacts and summary

```
1 name: CI
2
3 # Events that trigger workflow
4 on:
5   # Runs on all pushes to branches
6   push:
7   # Runs on all PRs
8   pull_request:
9   # Runs every day at midnight UTC
10  schedule:
11    - cron: "0 0 * * *"
12  # Manual Dispatch
13  workflow_dispatch:
14
15 jobs:
16   run-cace:
17     runs-on: ubuntu-22.04
18     name: Run CACE
19     timeout-minutes: 30
20     steps:
21       - uses: actions/checkout@v3
22         # Run CACE on the design
23       - name: Setup Environment and Run CACE
24         uses: efabless/cace-action@main
25         with:
26           pdk_family: 'sky130'
27           cace_root: '.'
28           cace_datasheet: 'cace/ota-5t.yaml'
29           cace_source: 'schematic'
30           token: ${{ secrets.GITHUB_TOKEN }}
31       # Upload artifacts
32       - name: Upload Artifacts
33         uses: actions/upload-artifact@v4
34         with:
35           name: runs
36           path: |
37             ${GITHUB_WORKSPACE}/runs/
38       # Add summary
39       - name: Add Markdown to Step Summary
40         run: cat runs/*/summary.md >> $GITHUB_STEP_SUMMARY
```

Run CACE summary

...

CACE Summary

general

- name: ota-5t
- description: A simple 5-transistor OTA
- commit: n/a
- PDK: sky130A
- cace_format: 5.0

authorship

- designer: Leo Moser
- company: Efabless
- creation_date: May 27, 2024
- license: Apache 2.0

netlist source: rcx

Electrical Parameters

Parameter	Testbench	Min Limit	Min Value	Typ Target	Typ Value	Max Limit	Max Value	Status
A0	ota-5t_tb.spice	50 V/V	61.48 V/V	any	81.97 V/V	any	94.6 V/V	Pass
UGF	ota-5t_tb.spice	1e6 Hz	1.247e+07 Hz	any	1.565e+07 Hz	any	1.812e+07 Hz	Pass
PM	ota-5t_tb.spice	60 °	86.02 °	any	86.31 °	any	86.86 °	Pass

Physical Parameters

Parameter	Tool	Min Limit	Min Value	Typ Target	Typ Value	Max Limit	Max Value	Status
Area	cace_area					600 μm^2	547 μm^2	Pass
LVS errors	cace_lvs						0	Pass
DRC errors	cace_drc						0	Pass

Job summary generated at run-time

More Examples

- sky130_ef_ip__opamp
- Operational amplifier
- More sophisticated example
- Several testbenches and plots

sky130_ef_ip_opamp Public

Unwatch 2 Fork 3 Star 4

main 1 Branch 0 Tags

Go to file Add file Code

About

Instrumentation amplifier (analog IP example)

Readme Apache-2.0 license Activity 4 stars 2 watching 3 forks Report repository

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No packages published Publish your first package

Contributors 2

RTimothyEdwards R. Timothy Edwards moles99 Leo Moser

Languages

Verilog 75.8% Tcl 24.2%

moles99 Merge pull request #1 from moles99/ci 30295bd · 2 hours ago 17 Commits

.github/workflows	Add CI and convert datasheet to YAML	9 hours ago
cace	Add CI and convert datasheet to YAML	9 hours ago
mag	Add CI and convert datasheet to YAML	9 hours ago
verilog	Add CI and convert datasheet to YAML	9 hours ago
xschem	Add CI and convert datasheet to YAML	9 hours ago
.gitignore	Add CI and convert datasheet to YAML	9 hours ago
LICENSE	Initial commit	4 months ago
README.md	Add CI and convert datasheet to YAML	9 hours ago
VERSION	Updated the characterization file in response to format c...	3 months ago

README Apache-2.0 license

Analog IP design example: Operational amplifier

This is a simple instrumentation amplifier demonstrating the use of the CACE system. It requires CACE to run the testbenches.

Schematics for this amplifier are done in xschem and can be found in the xschem/ directory. Layouts for this amplifier are done in mag and can be found in the mag/ directory.

The CACE specification is in cace/sky130_ef_ip__opamp.yaml. Testbench schematics are in the cace/templates/ directory. Note that testbenches are editable in xschem, and netlists can be generated, but netlists contain variables that must be substituted by CACE and cannot simulate prior to variable substitution by CACE.

To run CACE in GUI mode:

```
cace-gui cace/sky130_ef_ip__opamp.yaml
```

CACE can also be run from the command line, for example as:

```
cace cace/sky130_ef_ip__opamp.yaml --source schematic --parameter=parameter_name
```

- sky130_ef_ip__rdac3v_8bit
- 8-bit resistor ladder DAC with 3.3V output range
- Monte Carlo & mismatch analysis
- Dependency:
 - sky130_ef_ip__samplehold
- Postprocessing: GNU Octave

sky130_ef_ip__rdac3v_8bit Public

Unwatch 2 Fork 1 Star 1

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About

8-bit resistor ladder DAC with 3.3V output range

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Contributors 3

RTimothyEdwards azwefabless mole59 Leo Moser

Languages

MATLAB 67.7% Td 11.2% Verilog 8.4% Shell 7.4% Makefile 5.3%

RTimothyEdwards Revised the repository to make use of a Makefile and a ... 8793e13 · 2 hours ago 22 Comments

File	Description	Time
.github/workflows	Reduce iterations to 50, increase runtime to 60min	8 hours ago
cace	Reduce iterations to 50, increase runtime to 60min	8 hours ago
doc	Modified the DAC layout to accommodate changes mad...	last week
drc	Adding all mods for fixing verification issues in sche and ...	5 days ago
gds	Seeded repository with files from the local version.	4 months ago
lef	Seeded repository with files from the local version.	4 months ago
lvs	Modified the schematic to correct LVS property errors. O...	yesterday
mag	Revised the repository to make use of a Makefile and a ...	2 hours ago
verilog	Seeded repository with files from the local version.	4 months ago
xschem	Revised the repository to make use of a Makefile and a ...	2 hours ago
.gitignore	Revised the repository to make use of a Makefile and a ...	2 hours ago
LICENSE	Seeded repository with files from the local version.	4 months ago
Makefile	Revised the repository to make use of a Makefile and a ...	2 hours ago
README.md	Add CI and convert datasheet to YAML	9 hours ago

README Apache-2.0 license

Analog IP design example: 8-bit Resistive ladder DAC

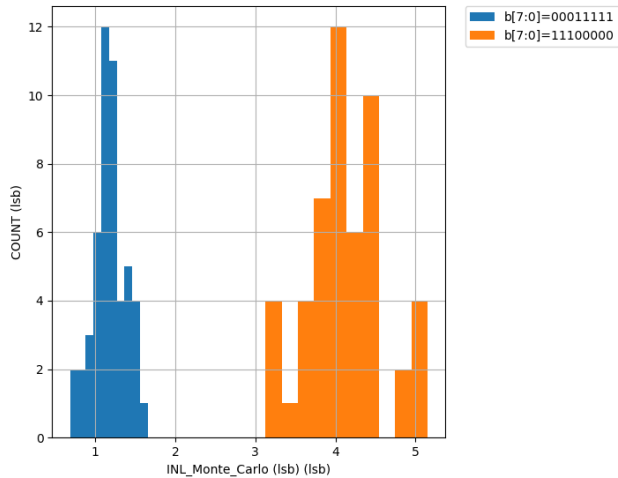
This is a simple 8-bit DAC demonstrating the use of the CACE system. It requires CACE to run the testbenches. Schematics for this DAC are done in xschem and can be found in the xschem directory. Layouts for this RDAC are done in magic and can be found in the mag directory.

The CACE specification is in cace/sky130_ef_ip__rdac3v_8bit.yaml. Testbench schematics are in the cace/templates directory. Note that testbenches are editable in xschem, and netlists can be generated, but netlists contain variables that must be substituted by CACE and cannot simulate prior to variable substitution by CACE.

To run CACE in GUI mode:

- Condition:
 - corner, seed and iteration
- .option
SEED=[{seed=12345} +
{iterations=0}]
- Collate the results
- Visualize in a plot

```
INL_Monte_Carlo:
description: Integral nonlinearity (INL), Monte Carlo
display: INL (Monte Carlo)
unit: lsb
spec:
  minimum:
    value: '-2'
    fail: true
    measurement: std3n-above
  typical:
    value: '0'
  maximum:
    value: '2'
    fail: true
    measurement: std3p-below
simulate:
  ngspice:
    template: voltage_output.spice
    collate: iterations
    format: ascii
    suffix: .data
    variables:
      - 'null'
      - result
  measure:
    - tool: octave
      filename: inl.m
  conditions:
    Vdvd:
      typical: '1.8'
    Vvdd:
      typical: '3.3'
    b[7:0]:
      maximum: '11100000'
      minimum: '00011111'
    temperature:
      typical: '27'
  iterations:
    description: Iterations to run
    display: Iterations
    minimum: '1'
    maximum: '50'
    step: linear
    stepsize: '1'
  corner:
    typical: mc
```



Success Story: Chipalooza

- 21 specified blocks to choose from
- 40+ proposals submitted
- 15 designs ready for layout in < 40 days
- 10 designs were on the test chip of the April shuttle

Designers were encouraged to use CACE...

- Of the 15 designs ready for layout, 12 used CACE
- Of 9 designs that passed schematic review but didn't make tapeout, 1 used CACE

Conclusion

- More support for different tools
 - Other simulators
 - KLayout DRC and LVS
- More mixed-signal simulation options
 - co-simulation: Verilator + ngspice
- More efficient simulation handling
- More complete handling of dependencies
- Better handling of search paths in xschem

Install [CACE](#) and read the [documentation](#).
Try out the [ota-5t](#) example design or create your own!

Thank You!